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COMPUTER SCIENCE III

WORKSHOP N° 2

1.2

- i) Identifier: $r' \backslash b [a-zA-Z] + [a-zA-Z|0-9|-]* \backslash b'$
- ii) Integer Literal: $r' \backslash b [0-9] + \backslash b'$
- iii) Floating Point Literal: $r' \backslash b [0-9] + \backslash b [0-9] + \backslash b'$
- iv) String Literal: $r' \backslash b \" . * ? ' \backslash b'$
- v) Single-Line Comment: $r' \backslash b // . * \backslash b'$
- vi) Multi-Line Comment: $r' \backslash b / * . * ? \backslash b / \backslash b'$
- vii) Whitespace: $r' \backslash b \backslash b +'$
- viii) Operators: $r' \backslash b [+ \backslash - * / = < > !] +'$
- ix) Keywords: $r' \backslash b (if | else | while | for | do | break | continue) \backslash b'$
- x) Hexadecimal Literal: $r' \backslash b 0x [0-9a-fA-F] +'$

2. a.

S
Program
Statement list
Statement Statement list
Assignment λ

Identifier "=" Expression ";" - - - - -
 λ | Term Expression'
| | Factor λ "+" Term Expression'
| | Number | Factor Term' λ
| | 5 | Number "*" Factor Term'
| | | 3 | Number λ
| | | | 2
 λ = 5 + 3 * 2 ;

b

S

Program

StatementList

Statement⁺ StatementList

ifStatement

λ

"if" "(" Expression ")" "{" StatementList "}" Elsepart

Term Expression'

Factor Term'

Identifier

X

En Expression' & Term' no están definidos
"<" & ">", tampoco en el
"context-free grammar"

No se puede generar
la expresión

"X > 0"

E.)

S

Program

Statement list

Statement Statement list

While statement λ

"While" "(" Expression ")" "{" Statement list "}"

Term Expression'

Statement Statement list

X Factor Term'

Assignment λ

Identifier Same error with "<"

Identifier "=" Expression ;

Term Expression'

Factor Term' "+" Term Expression'

Identifier λ

Factor λ

Number

1

while (x ...) { x = x + 1 ; }

d.

